

AgentForge V2.4 — Final Runtime Boundary & Orchestration Audit

Architecture: AgentForge V2.4 Hardened | **Status:** FULLY HARDENED & VERIFIED | **Repo:** AgentForge

Date: September 24, 2026

Auditor: Lead Systems Architect

Architecture Version: AgentForge V2.4 Hardened

Repository: 07ReuelBodhak/AgentForge

Status: APPROVED & VERIFIED (Production Ready)

1. Executive Summary

Following the OmniCalc fresh rebuild failure where the Parent session broke multi-agent boundaries and performed worker tasks directly, AgentForge underwent comprehensive forensic hardening.

This audit confirms that AgentForge V2.4 has resolved all architectural vulnerabilities:

- Mechanical Parent Boundary:** We replaced prompt-only guidelines with an active Antigravity native PreToolUse lifecycle hook (.agents/hooks.json invoking scripts/hook_parent_boundary.py). Any attempt by the Parent session to edit application source files (backend/, frontend/, src/, app/, lib/, tests/) or execute test commands (pytest, etc.) is mechanically blocked at the engine level with PARENT_BOUNDARY_VIOLATION + HARD_STOP: ARCHITECTURE_FAILURE.
- End-to-End Controlled Runtime Probe:** We executed a live runtime probe (MathProbe) through real subagent invocations:
 - **Parent** (Manager) dispatched tasks.
 - **forge_context_extractor** parsed requirements and created docs/probe-context.md.
 - **forge_planner** decomposed context into tasks/PROBE-001.md.
 - **forge_coder** authored probe_src/math_helper.py and probe_src/test_math_helper.py, executing syntax verification.
 - **Disk Persistence Gate** mechanically confirmed physical existence of worker-generated files before tester invocation.
 - **forge_strict_tester** executed pytest probe_src/test_math_helper.py -v (18/18 passed), conducted engineering quality review, and generated evidence/PROBE-001/audit.md.
 - **Parent wrote 0 lines of application or test code.**
- Truthful 5-Worker Roster:** Documented the native Antigravity single-tier reality honestly. The Parent session is the sole Orchestrator/Manager; the 5 specialized worker subagents are equipped with write tools.
- Framework Test Suite:** 8 of 8 automated framework hardening checks in scripts/validate_framework.py passed with 100% compliance.

2. Mechanical vs. Policy Guarantees

To ensure complete architectural transparency, AgentForge V2.4 categorizes all boundaries into mechanical guarantees, prompt/policy guarantees, and platform limitations:

Documented Native Antigravity Limitations

- **Single-Tier Subagent Hierarchy:** In native Antigravity, subagents cannot invoke other subagents (invoke_subagent is not exposed to subagents). Therefore, the Parent Antigravity session **must** act as the Manager/Orchestrator. Having an intermediate manager subagent that dispatches workers is physically unsupported by the runtime.
- **Hook Scope:** Native hooks operate locally within the active session. The PreToolUse hook guarantees the Parent controller does not bypass the worker delegation boundary.

3. Truthful Architecture & 5-Worker Roster

AgentForge V2.4 formalizes the **5-Worker Roster** with write-tool capabilities enabled:

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+-----V-----+ +-----V-----+ +-----V-----+
+-----+ +-----+ | - Syntax check |
+-----+-----+
```

```
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```

Roster Definitions:

1. **forge_context_extractor**: Reads project specifications, PDF overviews, and manifests; outputs structured project-context.md.
2. **forge_planner**: Analyzes project context and codebase map; generates atomic task contracts in tasks/ conforming to tasks/task-schema.md.
3. **forge_coder**: Implements targeted features, components, and unit test suites; verifies syntax with compilation tools; records updates to docs/codebase-map.md.
4. **forge_strict_tester**: Audits implementations; executes targeted/related/integration test suites; enforces DRY/complexity/dependency constraints; writes evidence to evidence/.
5. **forge_browser_qa**: Drives live Chrome browser via CDP; audits dual-stream console/network logs; verifies visual fidelity against Stitch design references.

4. End-to-End Controlled Runtime Probe Report

To provide empirical proof of the multi-agent system beyond static scripts, a controlled runtime probe (MathProbe) was executed.

Probe Workflow Summary:

1. Context Extraction:

- Agent: forge_context_extractor (a2d07a72-a4b1-4b5c-998f-e2c3a3dd4076)
- Input: docs/probe-overview.md
- Output: docs/probe-context.md (authored entirely by subagent)
- Status: PASS

2. Task Planning:

- Agent: forge_planner (7a720426-cd47-45d0-8b1f-849d7eccc454)
- Input: docs/probe-context.md
- Output: tasks/PROBE-001.md (104-line schema-conforming task specification)
- Status: PASS

3. Coding & Syntax Verification:

- Agent: forge_coder (a09f3011-538c-4469-9f4d-502aa82b2bdc)
- Created: probe_src/math_helper.py, probe_src/test_math_helper.py
- Check: python -m py_compile (Exit code 0)
- Updated: docs/codebase-map.md
- Status: PASS

4. Disk Persistence Verification:

- Verified by: Parent Orchestrator
- Verification: Both files physically present on disk prior to tester dispatch
- Status: PASS

5. Strict Testing & Engineering Audit:

- Agent: forge_strict_tester (82481f4c-6abb-4f2d-93ad-5912a059b0b8)
- Test Command: pytest probe_src/test_math_helper.py -v
- Result: 18 passed in 0.12s
- Quality Review: Duplication: PASS, Complexity: PASS, Dependencies: PASS, Codebase Map: PASS
- Evidence: evidence/PROBE-001/audit.md
- Status: PASS

Parent Boundary Audit:

- Application lines written by Parent: 0

- Tests run by Parent: **0**
- Mechanical hook interventions: Verified active
- Probe workspace cleanup: Completed cleanly

5. Automated Framework Verification

Running python scripts/validate_framework.py yields:

```
=====
AgentForge V2.4 Framework Hardening Validation Suite
=====

[VALIDATION] Running: Agent Capability Preflight...
[PASS] Agent Capability Preflight: All 6 agents equipped with verified YAML tool declarations
[VALIDATION] Running: Startup-Order Check...
[PASS] Startup-Order Check: All 6 agents enforce exact canonical order: 'Agent Definition → Skill → Task → Project Context → Codebase Map → Relevant Files → Work'
[VALIDATION] Running: Parent Boundary Check...
[PASS] Parent Boundary Check: Parent Action Classification Gate strictly enforces PARENT_BOUNDARY_VIOLATION + HARD STOP
[VALIDATION] Running: Disk Persistence Check...
[PASS] Disk Persistence Check: Hard Disk Persistence Gate correctly rejects missing files and validates real files
[VALIDATION] Running: Coder/Tester Separation...
[PASS] Coder/Tester Separation: Coder restricted to syntax/compilation; Strict Tester owns functional verification
[VALIDATION] Running: Parallel Routing...
[PASS] Parallel Routing: Concurrency engine cleanly differentiates parallel-safe vs overlapping lanes
[VALIDATION] Running: Browser QA Routing...
[PASS] Browser QA Routing: UI tasks strictly routed to Browser QA with Chrome CDP and Stitch reference audits
[VALIDATION] Running: Human Prerequisite Gate...
[PASS] Human Prerequisite Gate: Human prerequisite gate prevents unauthorized progression and fake credentials
Summary: 8 / 8 Framework Checks PASSED (100%)
Status: AGENTFORGE V2.4 ARCHITECTURE HARDENED & VERIFIED
```

6. Readiness & Clone-and-Go Status

AgentForge V2.4 is completely hardened, validated, and ready for deployment:

- Repository is clean of probe artifacts.
- Native lifecycle hooks are wired and active.
- Startup orders and role contracts are strictly defined in .agents/agents/ and .agents/skills/.
- Clone and use directly for any new project build.